Centrality-Based Epidemic Control in a Hospital Ward Contact Network

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STAT 636, Multivariate Analysis and Statistical Learning for Network Data, Texas A&M University, Fall 2026

1. Introduction

When an infection enters a hospital ward, only a few people can be isolated or kept away from work. The usual instinct is to isolate the people with the most contacts. A person can also matter because they link groups that rarely meet, even with fewer contacts. Network centrality measures formalize these ideas in different ways (Newman, 2018), and they often rank the same people differently, so it is not obvious which one should guide isolation.

Our primary question is **which centrality-based isolation rule most reduces the expected outbreak size in a recorded hospital contact network, and how its effectiveness changes when contact information is incomplete.** We compare degree, weighted betweenness, weighted eigenvector and harmonic centrality with random isolation, using a weighted SIR model simulated on a sensor-recorded ward network, and we use spectral communities to interpret the differences. We model preventive isolation before the infection arrives, so the results describe an idealized intervention, not a clinical policy.

2. Data and network construction

We use the SocioPatterns hospital ward dynamic contact network (Vanhems et al., 2013), downloaded from sociopatterns.org. Seventy-five people in a geriatric ward in Lyon, France, wore proximity sensors from December 6 to 10, 2010: 29 patients, 27 nurses and nurses' aides, 11 doctors and 8 administrative staff. Each of the 32,424 records marks a 20-second interval in which two people were within about 1.5 m of each other. Our file matched all of these published counts. The data are small enough for many repeated simulations, and the role labels give an external reference for detected communities.

We aggregate the records into a weighted, undirected network. An edge joins two people with at least one recorded contact, and its weight w_{ij} is their total contact hours divided by the 4.02 days covered by the records, so it is average daily contact hours. The network has 1,139 edges in one connected component, with density 0.41, mean degree 30.4, diameter 3, mean shortest path 1.6 steps and global clustering 0.59. Contact time is very uneven: median strength (total daily contact hours) is 0.61 and the maximum is 5.9. Doctors and nurses average 1.9 h/day, administrators 0.70 and patients 0.44 (Appendix Figure A1). Mixing follows the ward's organization: patients spend 75% of their contact time with nurses and 5% with other patients, and doctors spend 75% of theirs with other doctors. Figure 1a shows the network.

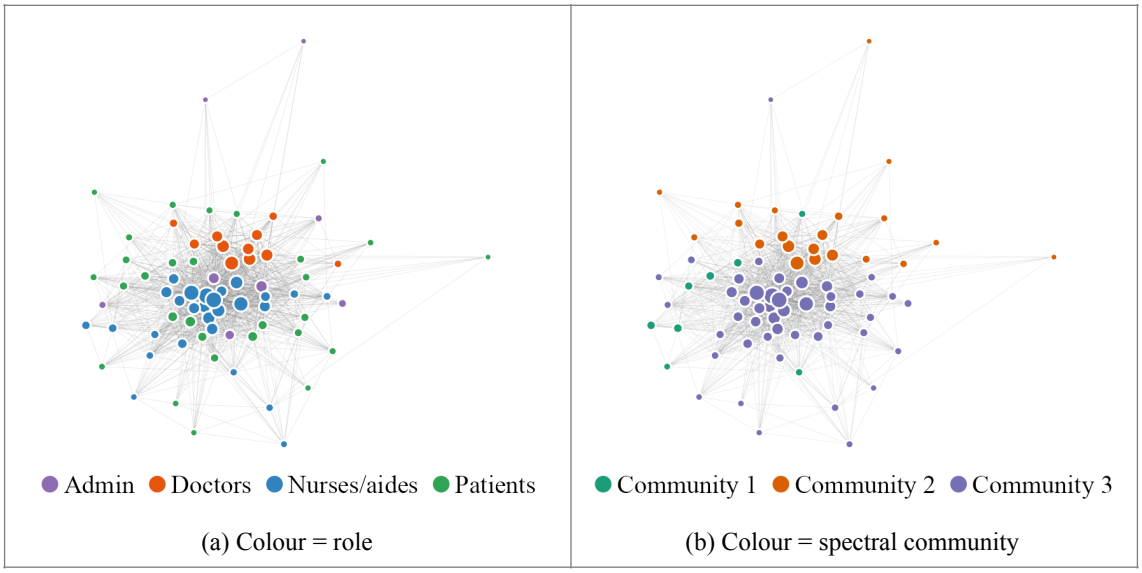


Figure 1. The ward contact network, drawn with the same layout in both panels. Node size shows daily contact hours and edge width shows contact time. Nurses form a dense core, doctors form a separate cluster, and many patients sit on the periphery. Panel (b) uses the same layout, so each person's role can be read from panel (a); it is discussed in Section 4.5.

3. Methods

3.1 Centrality and selection

We compute four measures with *igraph* (Csárdi and Nepusz, 2006). **Degree** counts distinct contacts, the simplest rule a ward could apply. **Eigenvector** centrality uses the contact weights and rewards contact time with people who are themselves central. **Betweenness** and **harmonic** centrality (Marchiori and Latora, 2000) use shortest paths with edge length $1/w_{ij}$, so long contact means a short distance. We use harmonic centrality instead of closeness because it stays defined when hidden edges disconnect the network. Each rule isolates its top k people (k = 1, 3, 5, 10), ranked once on the observed network, with ties broken by one fixed random vector. The baseline isolates k people chosen at random, with a new set drawn for every run. Rank correlations and the Jaccard overlap of the selected sets summarize agreement between rules.

3.2 Weighted SIR model

We wrote a discrete-time SIR simulator in R (R Core Team, 2025). One step is one day. An infected person i infects a susceptible contact j with probability $p_{ij} = 1 - \exp(-\lambda w_{ij})$, independently over contacts (Moreno et al., 2002). Person j is therefore infected on a given day with probability $1 - \exp(-\lambda \Sigma_i w_{ij})$, summing over people infected at the start of the day. This exponential form makes risk grow with contact time and never exceed one. People infected at the start of a day transmit and then recover with probability $\gamma = 0.2$ (a mean infectious period of 5 days); newly infected people transmit from the next day. Isolated people are removed before the epidemic and can neither be infected nor transmit. Each run starts from one person chosen uniformly among those not isolated and ends when no one is infected. The simulator passed eight checks against exact answers, including the one-day infection probability on a single edge (0.454 simulated vs 0.451 exact) and the full final-size distribution on a small weighted graph computed by brute-force enumeration (Appendix Table A1).

The transmission intensity λ (per contact hour) is not calibrated to a specific pathogen. We therefore chose three values from no-isolation pilot runs and fixed them before comparing any strategies: λ = 0.33, 0.75 and 2.0 give mean attack rates of about 25%, 50% and 75% (low, medium and high; Appendix Figure A2). The three levels cover a small, a moderate and a ward-wide outbreak.

3.3 Experiments and outcomes

The primary outcome is the **attack rate**: everyone ever infected, including the first case, divided by all 75 people. The secondary outcome is the probability of a major outbreak, defined in advance as at least 15 infected, with cutoffs of 10 and 20 as sensitivity checks. We report means with 95% Monte Carlo intervals (mean ± 1.96 SE). Strategies are simulated independently, so the interval for a difference uses the sum of the two squared standard errors. The proposal planned 500 runs per complete-data condition and allowed more if needed. We used **3,000**, because with 500 the standard error of a mean would be up to 1.6 percentage points, as large as the 1–3 point differences between centrality rules. To see which differences are stable, we repeated the comparison for k = 3, 5 and 10 with four more independent seeds and pooled the five replicates (15,000 runs per condition).

For **incomplete information**, we hid 20% or 40% of edges at random, computed centralities and chose whom to isolate on the reduced network, and then simulated spread on the full network, so hidden contacts still transmit. All nodes and retained weights are unchanged. We drew 30 missing-edge samples per level, shared by all strategies, with 100 runs per sample and condition. Intervals use the spread of the 30 sample means, so they include both epidemic and missing-data variation. For **community structure**, we applied spectral clustering with the normalized Laplacian $L = I - D^{-1/2}WD^{-1/2}$ (Ng et al., 2001; von Luxburg, 2007), chose K by the largest eigengap, checked $K \pm 1$, and compared communities with roles using the adjusted Rand index (ARI; Hubert and Arabie, 1985). Roles are an external reference, not ground truth. Finally, as the optional extension in our proposal, we generated 20 **degree-preserving rewired networks** (Maslov and Sneppen, 2002), permuted the observed weights at random over the new edges, and repeated the comparison at medium transmission (300 runs per network and condition). Appendix Table A2 lists every departure from the proposal.

4. Results

4.1 The measures disagree

Spearman rank correlations between the four measures range from 0.59 (betweenness vs eigenvector) to 0.95 (eigenvector vs harmonic; Appendix Figure A3). The first pick differs: degree selects an administrator with 61 distinct contacts but only 2.0 contact hours per day, while the three weighted measures select nurses with heavy

contact. Betweenness also selects two doctors and one patient in its top ten, whereas eigenvector and harmonic choose almost the same people (Jaccard overlap 0.67–1.0 for $k = 1$ to 10).

4.2 Every centrality rule beats random isolation

Figure 2 and Table 1 show the main comparison. For $k \geq 3$, every centrality rule gives smaller outbreaks than random isolation at all three intensities, and every interval for the difference excludes zero (Appendix Figure A4). Isolating ten people lowers the attack rate from 17.5% (random) to 4.8–7.0% at low transmission and from 38.6% to 19.4–24.4% at medium transmission, 37–73% fewer infections. At high transmission it falls from 61.2% to 49.7–52.8%. Isolating a single person changes the attack rate by at most 3.1 percentage points. Random isolation helps less because about half of the ward are patients or administrators, who have little contact time.

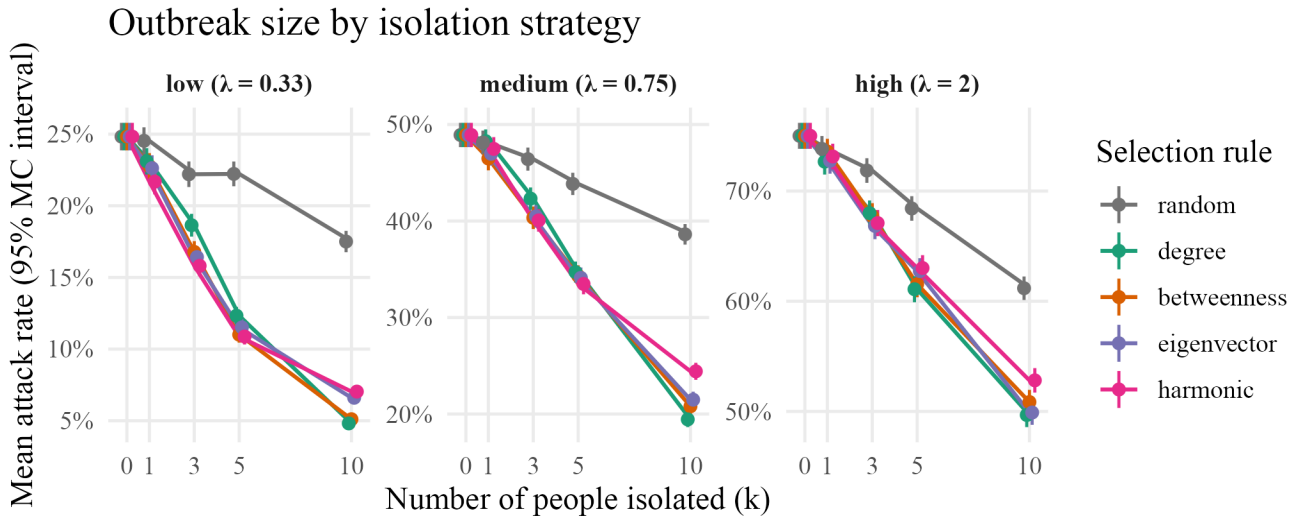


Figure 2. Mean attack rate against the number of people isolated, by selection rule, at three transmission intensities. Each point is the mean of 3,000 simulated outbreaks; bars are 95% Monte Carlo intervals. Points at $k = 0$ are the no-isolation baseline.

Table 1. Attack rate, % (95% Monte Carlo interval), for no isolation and for isolating $k = 3$ or 10 people; 3,000 runs per cell. Low, medium and high are $\lambda = 0.33$, 0.75 and 2.0.

Rule	Low, k = 3	Low, k = 10	Medium, k = 3	Medium, k = 10	High, k = 3	High, k = 10
No isolation	24.8 (23.9–25.8)	24.8 (23.9–25.8)	48.9 (47.7–50.2)	48.9 (47.7–50.2)	75.0 (73.8–76.2)	75.0 (73.8–76.2)
Random	22.2 (21.3–23.1)	17.5 (16.8–18.3)	46.4 (45.2–47.6)	38.6 (37.6–39.7)	71.9 (70.7–73.0)	61.2 (60.1–62.2)
Degree	18.6 (17.9–19.4)	4.8 (4.6–5.1)	42.3 (41.2–43.5)	19.4 (18.6–20.2)	68.0 (66.8–69.1)	49.7 (48.6–50.8)
Betweenness	16.8 (16.0–17.5)	5.1 (4.8–5.4)	40.3 (39.2–41.5)	20.8 (20.0–21.6)	67.7 (66.6–68.9)	50.9 (49.7–52.0)
Eigenvector	16.4 (15.7–17.2)	6.6 (6.2–7.0)	40.5 (39.3–41.6)	21.5 (20.7–22.3)	66.8 (65.6–68.0)	49.9 (48.8–51.0)
Harmonic	15.8 (15.1–16.5)	7.0 (6.7–7.4)	40.1 (38.9–41.2)	24.4 (23.5–25.3)	67.1 (65.9–68.3)	52.8 (51.7–53.9)

4.3 The best rule depends on isolation capacity

Figure 3 compares each weighted measure with degree using the pooled 15,000 runs. Two patterns held in at least four of the five replicates. With few isolation slots ($k = 3$ or 5) at low and medium transmission, all three weighted measures beat degree by 0.8–3.0 percentage points, and all twelve intervals exclude zero. Degree spends one of its few picks on the administrator who meets many people briefly. At $k = 10$ the order reverses: degree has the lowest attack rate at every intensity, eigenvector is 1.2–1.6 points worse and harmonic 2.3–4.6 points worse, and betweenness is 0.1–1.7 points worse. At high transmission and $k = 3$ or 5 the weighted measures are within 0.8 points of degree, and five of the six intervals include zero. Differences below about one point can change sign between replicates. For example, harmonic minus degree at high transmission and $k = 3$ was −0.9 points in the first replicate and +0.2 points pooled, so we interpret only the consistent patterns. The probability of at least 15 infections ranks the rules almost the same way (Appendix Figure A5), and at medium transmission it barely changes when the cutoff is 10 or 20 instead (Appendix Figure A6).

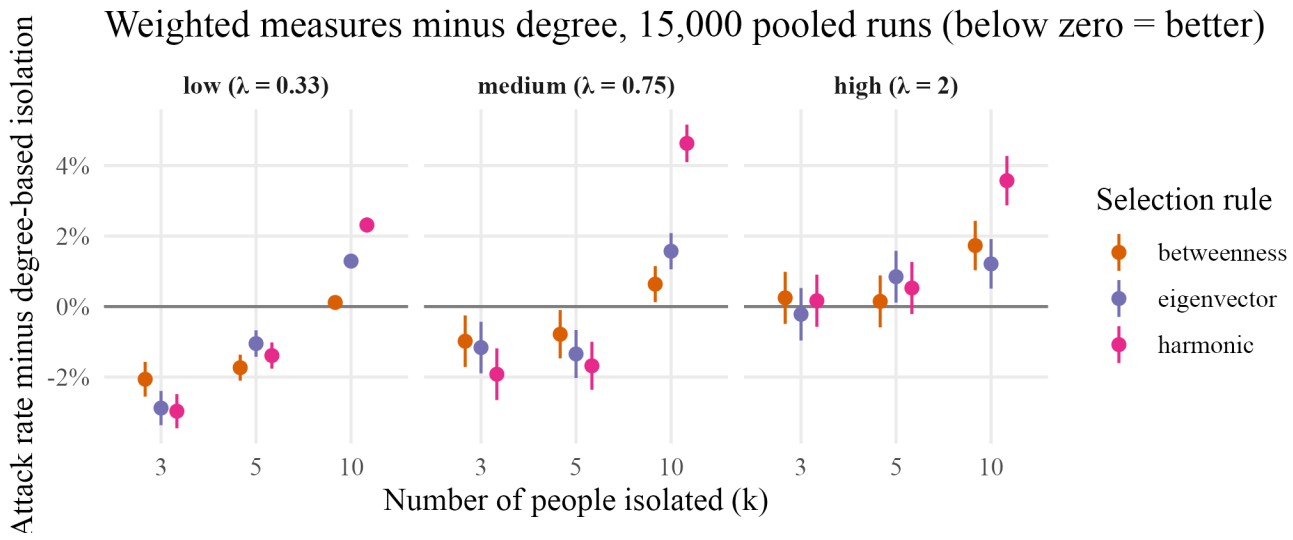


Figure 3. Attack rate of each weighted measure minus that of degree-based isolation, pooled over five independent replicates (15,000 runs per condition), with 95% Monte Carlo intervals for the difference. Below zero means better than degree.

4.4 Incomplete contact information

Hiding contacts changes who is selected but hardly changes how well isolation works (Figure 4). The mean overlap between the sets chosen on incomplete and complete data ranges from 30% to 92% (Appendix Figure A7). Attack rates, however, change by −2.8 to +3.1 percentage points, and for $k \geq 3$ every centrality rule remains better than random isolation at both 20% and 40% missingness (72 of 72 paired intervals exclude zero). The largest loss is for degree at medium transmission and $k = 10$, from 19.4% to 22.5% at 40% missingness. Our interpretation is that the ward has many similar high-contact nurses, so missing a few contacts swaps one busy nurse for another of similar importance.

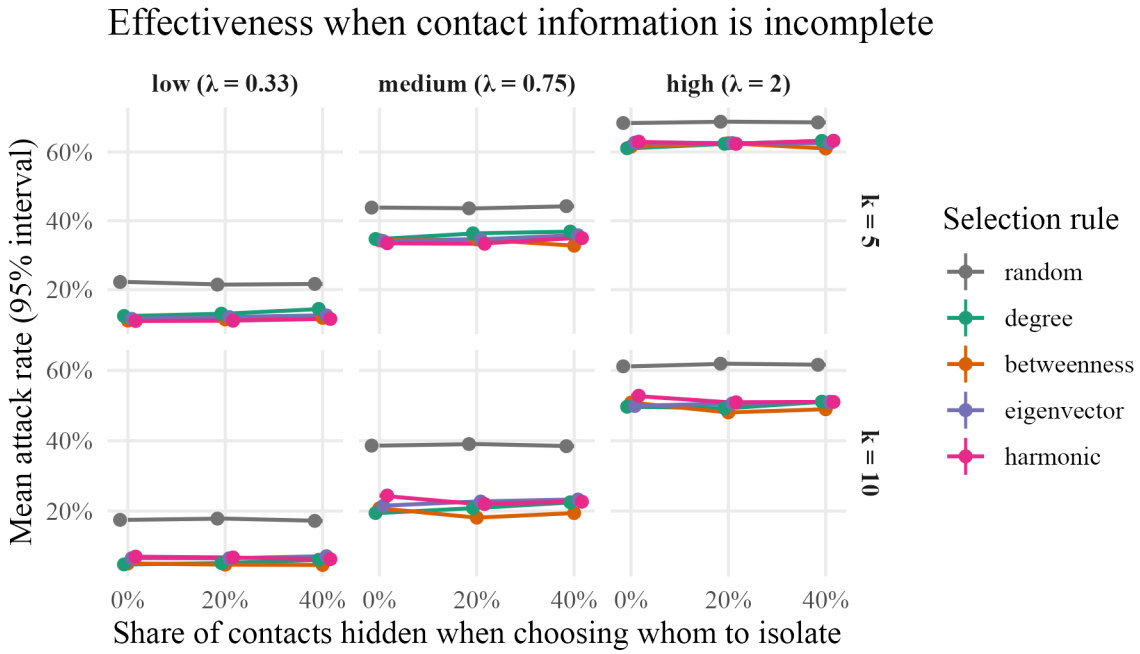


Figure 4. Mean attack rate when 0%, 20% or 40% of contacts are hidden while ranking (spread is always simulated on the full network), for $k = 5$ and 10. At 20% and 40%, points are means over 30 missing-edge samples × 100 runs, and the 95% intervals use the spread of the sample means, so they include both epidemic and missing-data variation. Points at 0% are the complete-data results (3,000 runs).

4.5 Community structure and rewiring

The eigengap selects $K = 3$ communities (Figure 1b; Appendix Figure A8): all 11 doctors with 8 patients and 2 administrators; a large group of 24 nurses, 6 administrators and 16 patients; and a small group of 3 nurses and 5

patients. They mix roles, and the ARI with roles is only 0.17–0.22 for $K = 2$ to 4. Every person in the top ten of eigenvector or harmonic centrality lies in the large nurse group, and the people these two rules select spend on average 5–10% of their contact time outside their own community (9–10% for the top ten), compared with 24% for the average person. The top ten of degree and betweenness reach two communities. This is consistent with the weighted measures falling behind degree at $k = 10$ by isolating people who overlap in whom they meet, although we did not test that mechanism directly. In the rewired networks the K = 3 partition has no community structure left (modularity −0.02, against 0.28 in the ward), although 62% of edges stay in place. With the same isolation sizes, outbreaks are much larger (58–64% vs 40–46% at $k = 3$ across all five rules), and degree is best or tied at every k (Figure 5). The small-k advantage of the weighted measures is therefore not a consequence of the degree sequence alone. Because rewiring also reshuffles the weights, it does not show that community structure by itself is responsible.

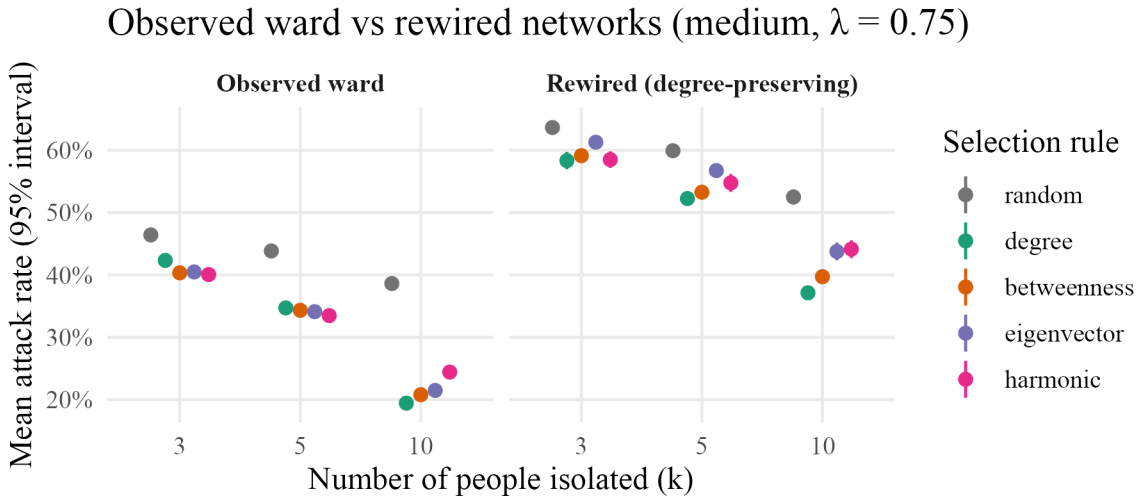


Figure 5. Mean attack rate at medium transmission (λ = 0.75) in the observed ward (3,000 runs per point) and in 20 degree-preserving rewired networks with randomly permuted weights (300 runs per network; interval from the spread across the 20 networks). Bars are 95% intervals.

5. Discussion

In this ward, any centrality-based rule is much better than random isolation, but no single measure is best in every condition. With very limited capacity, measures that use contact time (betweenness, eigenvector, harmonic) are better. Once about ten people can be isolated, plain degree does as well or better. Incomplete contact data changes the selected people substantially but has little effect on outcomes.

Limitations. The data cover one ward over four days, and 4 staff members and 2 patients did not take part (Vanhems et al., 2013). The network is static, so the timing and order of contacts are ignored. The SIR model and its λ values are illustrative rather than calibrated. Isolation is preventive, and isolated staff are not replaced. Hidden edges are removed at random, which is a simplified missing-data pattern. Monte Carlo intervals reflect variation within these assumptions, not generalizability to other hospitals.

Open questions and how to address them. (i) Top-k rules may pick people who meet the same contacts. A greedy rule that adds, one at a time, the person whose isolation most lowers the simulated attack rate given those already chosen would test this. (ii) A temporal SIR run on the time-stamped records would show whether aggregating to a static network changes the ranking of rules. (iii) Real missing data are unlikely to be random: brief contacts are forgotten first. Hiding the lowest-weight edges first, or hiding contacts of patients more often, would test this. (iv) Adaptive re-ranking after each removal and strength (total contact hours) as a fifth rule are natural extensions. Repeating the analysis on other SocioPatterns data sets would show how far the conclusions generalize.

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SocioPatterns collaboration. Hospital ward dynamic contact network. https://sociopatterns.org/datasets/hospital-ward-dynamic-contact-network/ (file hospital_lyon_contacts.dat.gz, accessed November 3, 2026). Distributed under the Creative Commons Attribution-NonCommercial-ShareAlike 3.0 licence (CC BY-NC-SA 3.0). We acknowledge the SocioPatterns collaboration (www.sociopatterns.org) for making the data available.

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AI disclosure statement

We used Claude (Anthropic), an AI assistant, through the Claude app and the Claude Code command-line tool. Tasks performed with AI: (1) reviewing the course requirements and helping draft and format the project proposal; (2) writing the R code for downloading the data, building the network, computing centralities, the SIR simulator and its checks, all experiments, community detection and figures; (3) running the analysis; (4) drafting the presentation slides, speaker notes and this report; and (5) auditing the project before submission: re-running the code from a clean copy, reviewing it for errors (this found and fixed an error in the role-mixing shares), adding simulator checks, checking every number in the report and slides against the code output, and checking each reference against its publisher or DOI record; and (6) preparing the final files: setting Times New Roman in every figure and document, building the PowerPoint deck from the slide sources, and writing the speaker script, question preparation and submission checklist. The extent of use was substantial for code, first drafts and checking. The team chose the research question and the methods in the proposal, divided ownership of the analysis as listed under Authors' contributions, reviewed the code, every figure and the results, and edited the slides and this report. Each member can explain all material in the presentation and report.

Authors' contributions

Each member owned one part of the analysis, about a third of the report and about a third of the talk, and all three reviewed the final report and slides.

- **Arvinder Singh Mundra:** project coordination and framing: the research question and motivation, data download and network construction, descriptive statistics and presentation figures (R scripts 00–02 and 10, run_all.R); Sections 1, 2 and 5 of the report and its final assembly; presented slides 1–3 and the conclusion; code repository and Canvas submissions.
- **Hasini Kavuru:** methods and the main experiment: the four centrality measures and selection rules, the weighted SIR simulator and its checks, the choice of transmission intensities and the complete-data isolation

experiment (R scripts 03–06 and functions.R); Sections 3, 4.1 and 4.2 and Table 1; presented slides 4–6 and the open questions; rehearsal timing and the check of numbers against the code output.

- **Satvika Reddy Akati:** robustness and extensions: the comparison across isolation capacities, the incomplete-data experiment, spectral communities, the replicate seeds and the rewiring extension (R scripts 07–09); Sections 4.3–4.5 and the Appendix; presented slides 7–9; the reproducibility run from a clean copy and the check of references.

Speaking split in the presentation: Arvinder Singh Mundra, slides 1–3 and the end of slide 10 (about 4.2 minutes); Hasini Kavuru, slides 4–6 and the start of slide 10 (about 4.5 minutes); Satvika Reddy Akati, slides 7–9 (about 4.0 minutes).

Appendix

Code and reproducibility. All scripts are in the attached file "17 - Centrality-Based Epidemic Control in a Hospital Ward Contact Network - Code.zip". Open STAT636_Group17.Rproj in RStudio and source run_all.R, which takes about 5 minutes on a laptop. The contact file is downloaded from SocioPatterns on the first run and checked against the published counts and the MD5 checksum of the file used here. Every random step is seeded, and a rerun from a clean copy reproduced every table used in this report.

Table A1. Checks of the SIR simulator against exact answers (all passed; table T0_simulator_checks.csv in output/tables).

Check	Result
No transmission ($\lambda = 0$)	lambda = 0 gives exactly one case
One-day infection probability on one edge ($w = 2$, $\lambda = 0.3$)	empirical 0.4542 vs exact 0.4512
Mean infectious period ($\gamma = 0.2$)	mean infectious days 5.04 (expected 5)
Isolated person (4-node complete graph, $\lambda = 2$)	isolated node infected in 0 of 300 runs; other three infected in 300 of 300
Two-person final size (closed form)	P(second infected) simulated 0.8024 vs exact 0.8043
Final-size distribution, 4-node weighted graph (exact enumeration)	mean final size simulated 3.5319 vs exact 3.5389; max |P sim - P exact| over sizes 1-4 = 0.0035; enumerator reproduces check 5 (0.8043)
Only one person not isolated	with 3 of 4 isolated, every run has exactly one case (the remaining person)
First case and random isolation sets	fixed sets: seeds never isolated, chi-square p = 0.937; random sets: 120 distinct sets in 500 runs, seed outside set in all runs

Table A2. Departures from the proposal and implementation choices the proposal left open.

Item	What we did and why
Runs per complete-data condition	3,000 instead of the planned 500, as the proposal allowed. With 500 runs the intervals were too wide to separate the centrality rules.
Replicate seeds (added)	Four extra independent replicates for k = 3, 5, 10 (15,000 pooled runs), to check which differences between rules are stable.
Pilot targets for λ	The proposal said only "three intensities from no-isolation pilot runs". We targeted mean attack rates of 25%, 50% and 75% (40 grid values × 1,000 runs, loess fit), and confirmed them with 4,000 fresh runs.
Missing-data replication	As planned (30 samples × 100 runs). Replication was not increased, because the changes from complete data were small relative to the differences from random isolation.
Rewiring extension	Done (optional in the proposal): 20 networks, weights permuted at random over the new edges, medium transmission only, k = 3, 5, 10.
Not done	Adaptive rankings and other missing-data mechanisms (extensions in the proposal; listed as open questions).
Presentation length	The proposal planned 15 minutes; the instructor set 12–13 minutes. The report follows the instructor's 5-page limit including main figures and tables.

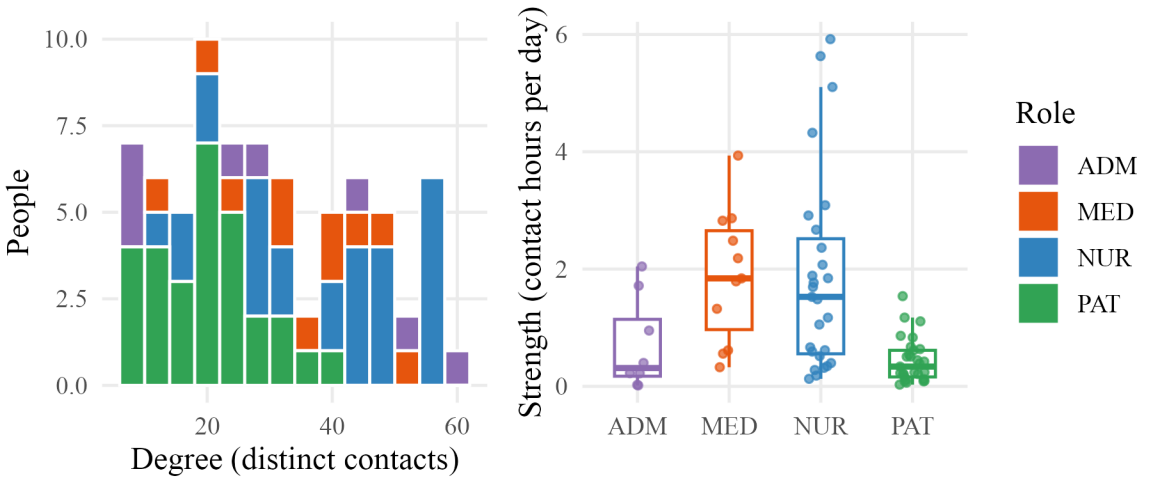


Figure A1. Degree (left) and daily contact hours, or strength (right), by role; each point is one person. Nurses and doctors have far more contact time than patients and administrators, which is why the weighted measures favour nurses (Sections 2 and 4.1).

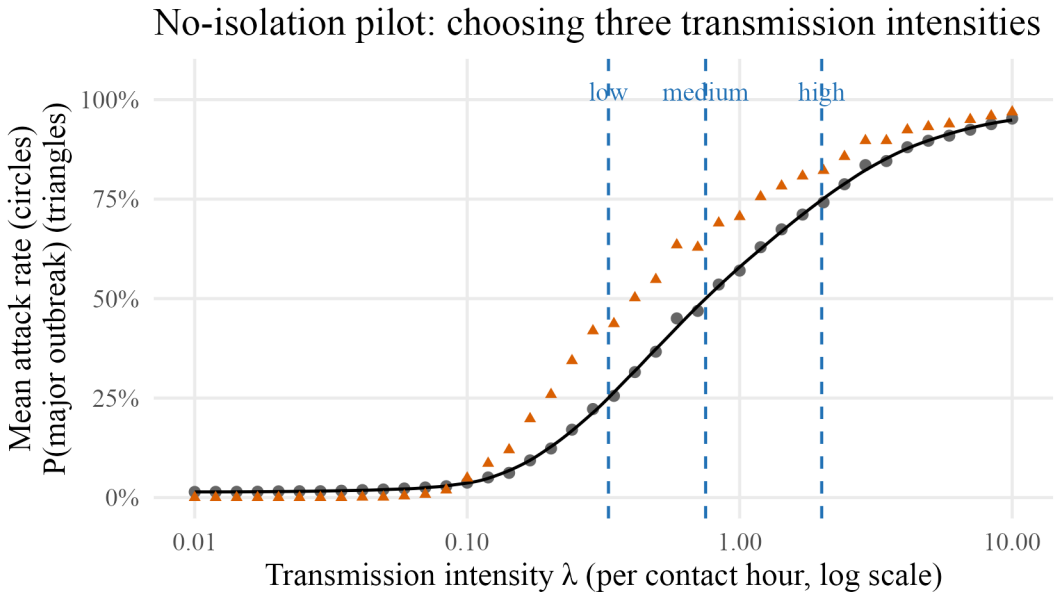


Figure A2. No-isolation pilot used to fix the three transmission intensities (dashed lines). Circles: mean attack rate (1,000 runs per grid value) with the fitted curve; triangles: probability of at least 15 infections. At the low value only 44% of outbreaks reach 15 infections, and those average 55% of the ward, so the mean of 25% mixes early die-outs with large outbreaks (Section 3.2).

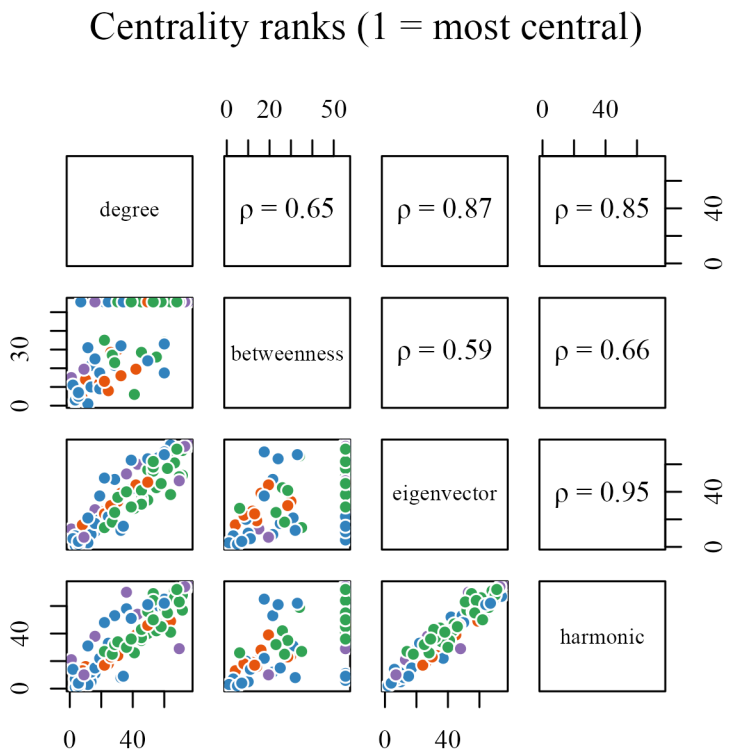


Figure A3. Pairwise centrality ranks (1 = most central; tied scores share their average rank), coloured by role, with Spearman correlations (as in Section 4.1). Eigenvector and harmonic nearly coincide, and betweenness differs most from the others (Section 4.1).

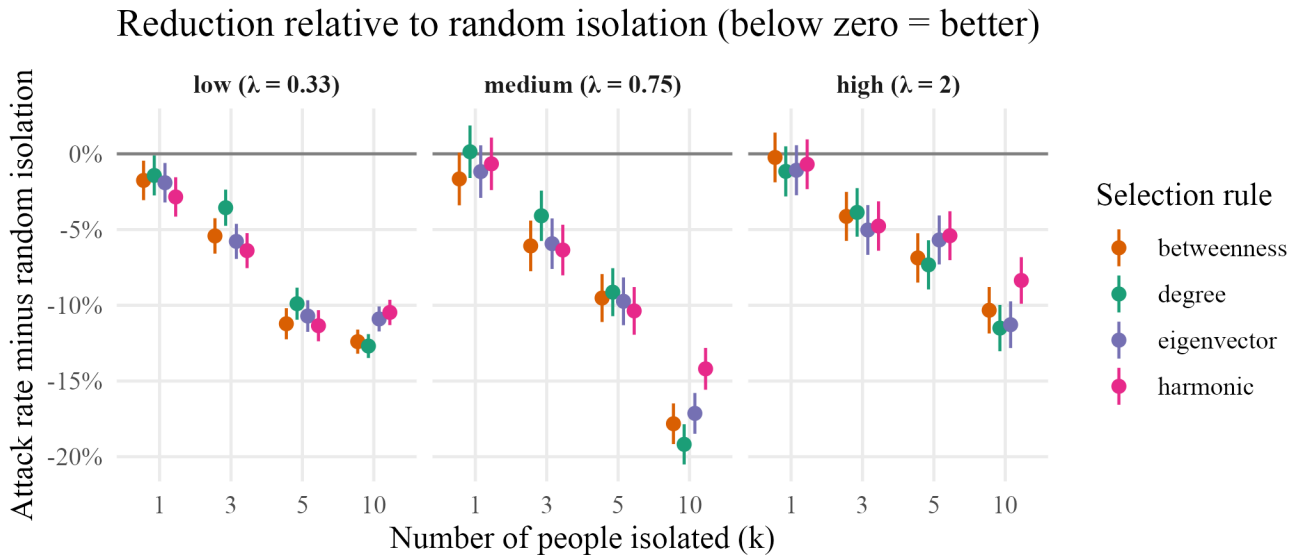


Figure A4. Attack rate of each centrality rule minus random isolation, with 95% Monte Carlo intervals for the difference (3,000 runs per rule and condition). Every interval is below zero for $k \geq 3$ at every intensity (Section 4.2).

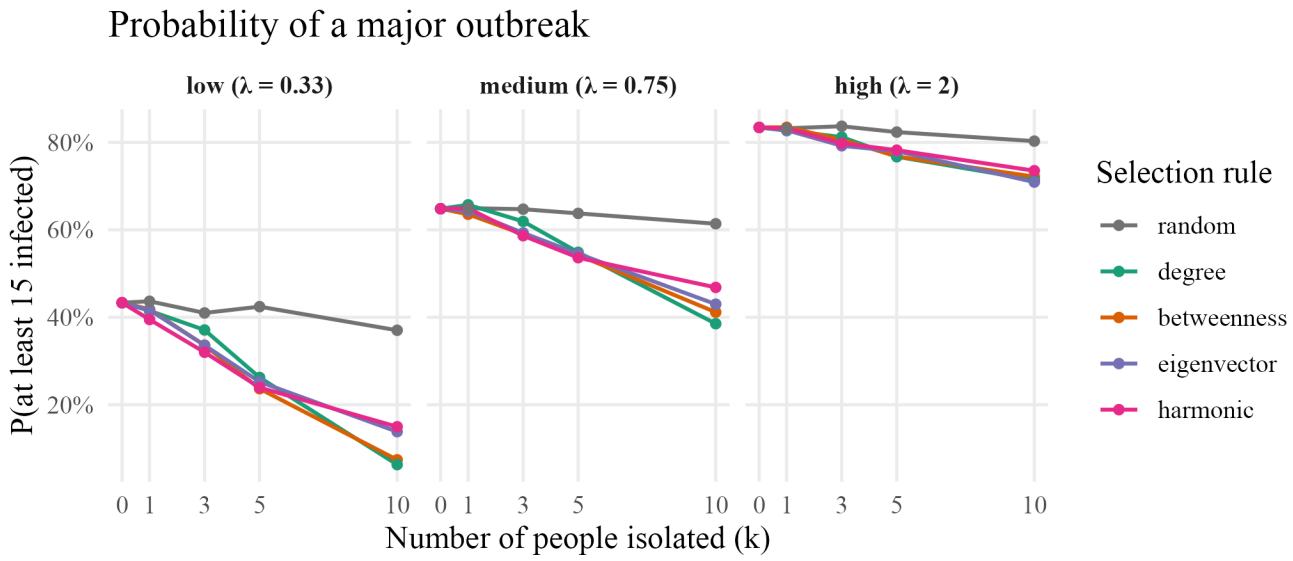


Figure A5. Probability of at least 15 infections by rule and intensity (3,000 runs per point; binomial standard errors at most 0.01). At high transmission a major outbreak remains likely even with ten people isolated (Section 4.3).

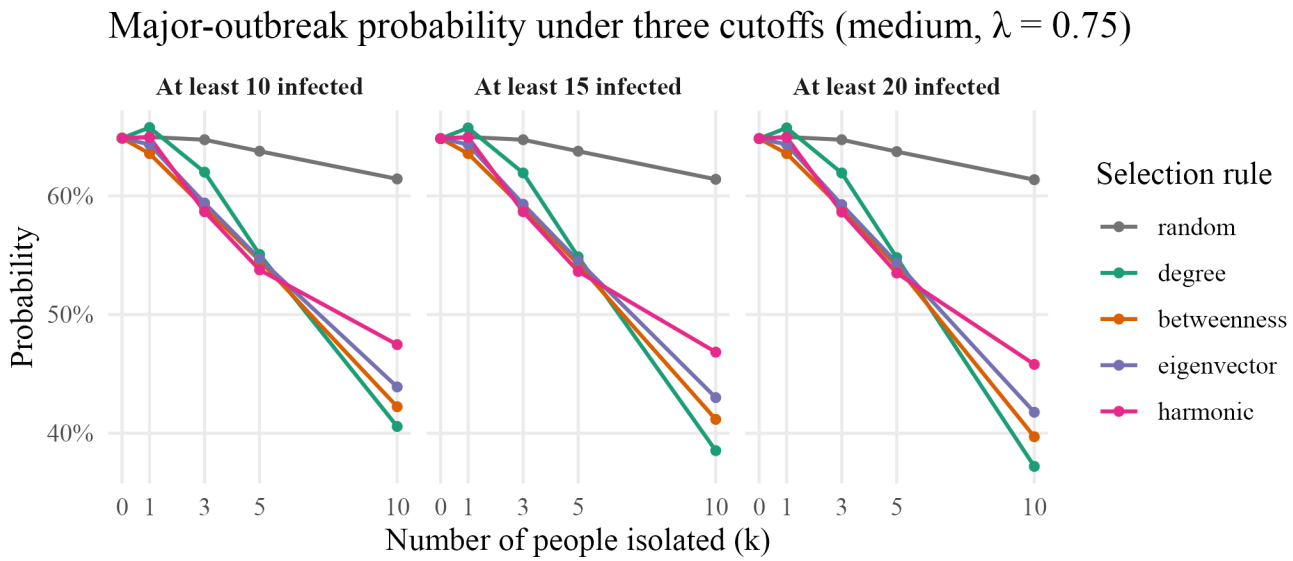


Figure A6. Probability of a major outbreak at medium transmission for cutoffs of 10, 15 and 20 infections (3,000 runs per point). Few outbreaks end with 10 to 19 cases (at most 3.4% of runs in any condition), so the three panels are nearly identical (Section 4.3).

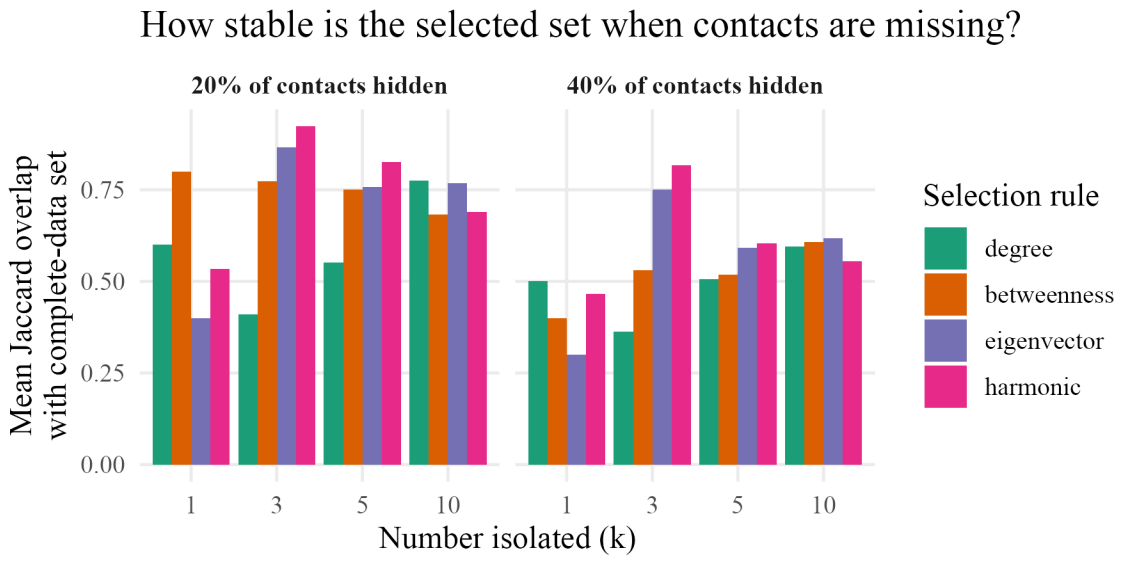


Figure A7. Mean Jaccard overlap between the people selected on incomplete data and on complete data, over 30 missing-edge samples per level. Selections change most for k = 1 and for degree at k = 3 (Section 4.4).

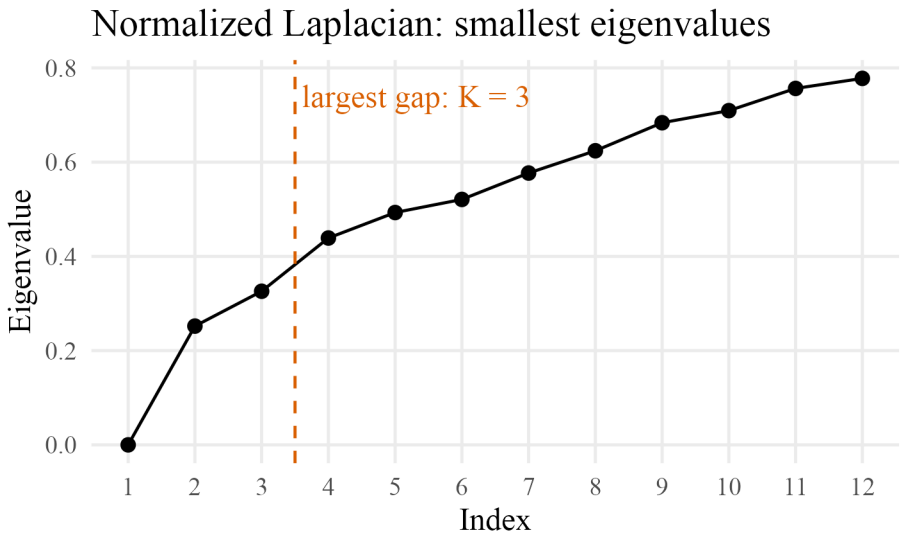


Figure A8. The twelve smallest eigenvalues of the normalized Laplacian. The largest gap after the first eigenvalue gives K = 3; K = 2 and 4 give a similar ARI with roles (0.22 and 0.17) (Section 4.5).